

# Adrenal Insufficiency — Clinical One-Pager

Localize the level · diagnose with AM cortisol & stim test · treat crisis empirically. Endocrine Society 2016 / ESE-ES 2024. Adapted from TeachIM.org (CC BY-NC 4.0) — clinician-reviewed.

## Three levels

| Level     | Lesion                | ACTH                    | Aldo/<br>RAAS |
|-----------|-----------------------|-------------------------|---------------|
| Primary   | Adrenal               | ↑ (hyperpigment)        | Lost          |
| Secondary | Pituitary             | low/nl                  | Intact        |
| Tertiary  | Hypothal / exog<br>GC | low (even post-<br>CRH) | Intact        |

**Primary AI loses aldosterone → hypovolemia, ↑K, ↓Na → crisis-prone. Central spares RAAS.**

## Causes

- **Primary:** autoimmune adrenalitis (most common); infection (TB, fungi, syphilis); hemorrhage; mets/tumor; drugs.
- **Central:** exogenous glucocorticoids (**most common cause overall**); mass/craniopharyngioma/mets; XRT; infiltrative (hemochromatosis, sarcoid); Sheehan; surgery.
- **Drugs:** steroids, azoles, etomidate, mitotane, checkpoint inhibitors; rifampin/phenytoin/PBT (↑metabolism).

## Diagnosis

- **1st test: 8 AM serum cortisol.** Very low rules in; clearly high rules out; indeterminate → stim test.
- Board cutoffs (older assay): <3 ruled-in; <10 suspected; >18 ruled-out.
- **Cutoffs are ASSAY-SPECIFIC:** modern monoclonal/LC-MS-MS read ~20-30% lower → rule-out / stim peak ~**14-15 µg/dL**, not 18.
- **ACTH (cosyntropin) stim:** 250 µg IV/IM, cortisol at 0 + 30-60 min; subnormal peak = AI.
- **ACTH level localizes:** ↑ = primary; low/nl = central. Then 21-OH-Ab (primary) or pituitary MRI (central).

## Chronic replacement

| Type    | Regimen                                                                                    |
|---------|--------------------------------------------------------------------------------------------|
| Primary | Hydrocortisone 20-25 mg/24h split (AM-weighted) + <b>fludrocortisone 0.1 mg qAM</b> ± DHEA |
| Central | Hydrocortisone 15-20 mg/24h split; <b>no fludrocortisone</b> (RAAS intact)                 |

At stress doses (HC >~50 mg/d) mineralocorticoid effect is adequate — fludro can be held.

## Adrenal crisis — treat NOW

- Presents: hypotension/shock ± fever, abd pain, N/V, fatigue; ± ↑K / ↓Na / hypoglycemia.
- **Hydrocortisone 100 mg IV bolus → 200 mg/24h (or 50 mg q6h) + aggressive IV fluids.** Treat empirically — do NOT await testing.
- If a stim test is still pending, use **dexamethasone** (no cortisol-assay cross-reaction).

## Patient safety / sick-day rules

- Double/triple the oral dose for febrile illness or stress; IV hydrocortisone for major stress/surgery.
- Every AI patient: emergency IM hydrocortisone kit + medical-alert ID.
- Under-replacement during stress is what precipitates crisis.